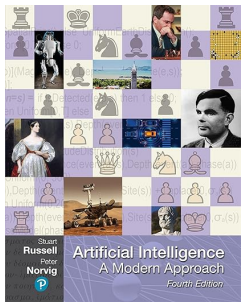
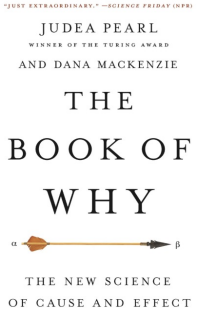
Lesson 08.4:

Instructor: Dr. GP Saggese, gsaggese@umd.edu

References:

- AIMA (Artificial Intelligence: a Modern Approach)
- Pearl et al., The Book of Why, 2017
- Facuce





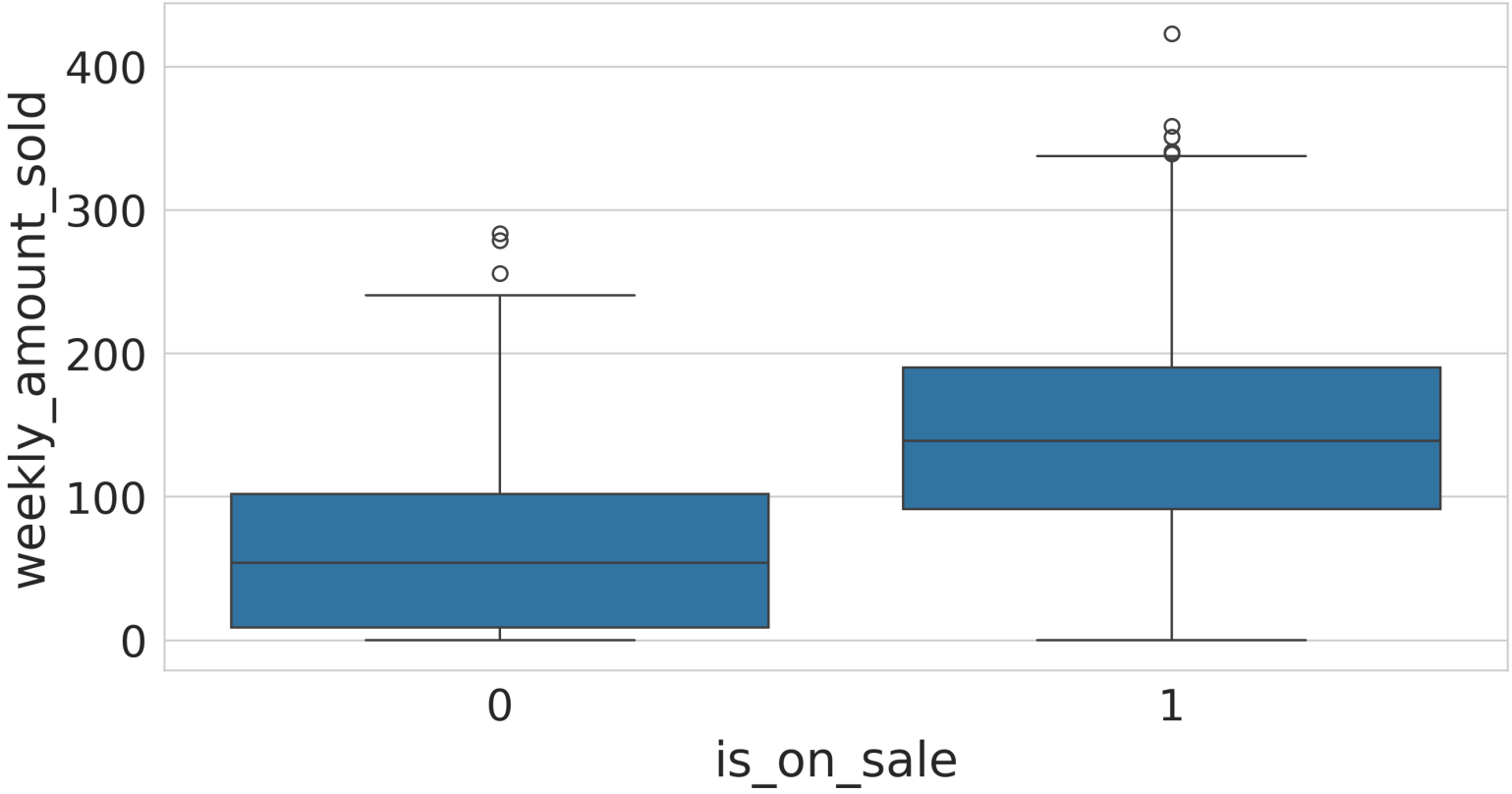
Definition

- Causal inference question: *"what is the impact of T on Y?"*
  - E.g., `is_on_sale` $\rightarrow$ `on_weekly_amount_sold`
- **Treatment for unit i** $T_i = 1$ if unit i has received the treatment
  - E.g., `(weeks_to_xmas, is_on_sale)` = "apply price drop on a specific week"
  - Sometimes T is denoted with D
- **Outcome** is the variable one wants to influence, Y_i
  - E.g., `weekly_amount_sold`
- **Unit of analysis** is the thing you wish to intervene on / treat, e.g.,
  - *"People who buy Y"*
  - *"When is the best moment to have sales"*

	store	weeks_to_xmas	avg_week_sales	is_on_sale	weekly_amount_sold
0	1	3	12.98	1	219.60
1	1	2	12.98	1	184.70
2	1	1	12.98	1	145.75
3	1	0	12.98	0	102.45
4	2	3	19.92	0	103.22
5	2	2	19.92	0	53.73

The Fundamental Problem of Causal Inference

- *"You can never observe the same unit with and without treatment"*
- **Example**
  - Plot outcome `weekly_amount_sold` by treatment `is_on_sale`
  - Stores with price drops sell more (from 50 to 150)



  - Matches intuition: *"people buy more when prices are low due to sales"*
  - Magnitude of change seems too large

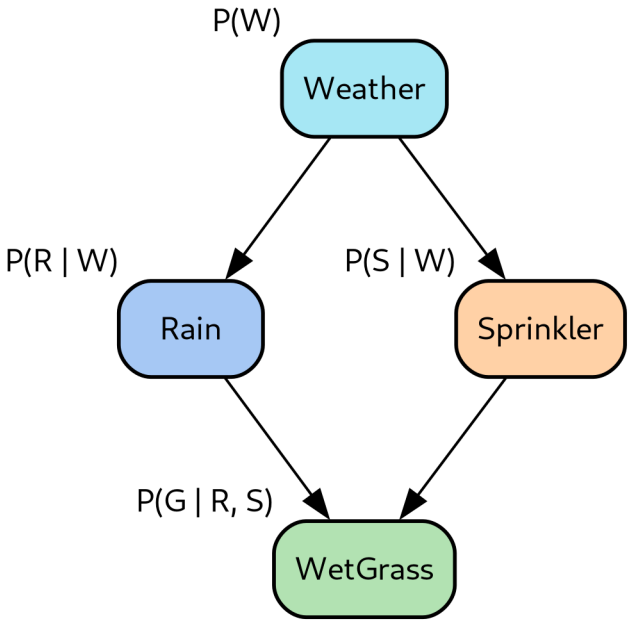
Causal Model

- A **causal model** is a series of assignments $\leftarrow$
  - Causality is non-reversible; you can't rearrange the equation
- **Example**

$$T \leftarrow f_t(u_t)$$
$$Y \leftarrow f_y(T, u_y)$$

where:

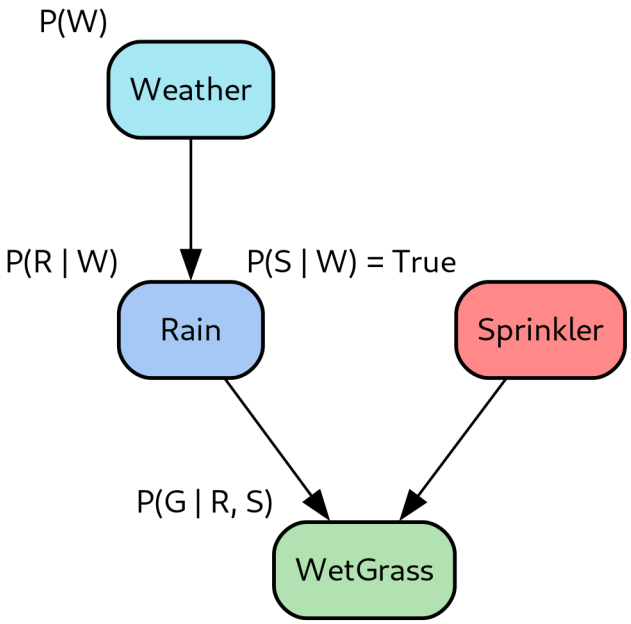
- u_t, u_y are exogenous variables (outside the model)
  - Account for everything not covered by model variables
- Treatment T and u_y cause outcome Y
- T and Y are endogenous to the model



Causal Model: Sales Example

- Causal model for sales example

$$IsOnSales \leftarrow f_t(u_t)$$
$$AmountSold \leftarrow f_y(IsOnSales, u_y)$$



- You could add a *BusinessSize* in the model

$$BusinessSize \leftarrow f_s(u_s)$$
$$IsOnSales \leftarrow f_t(BusinessSize, u_t)$$
$$AmountSold \leftarrow f_y(IsOnSales, BusinessSize, u_y)$$

- One can make an assumption of linear model for $f_y(.)$ and say

$$AmountSold = \alpha + \beta_1 IsOnSales + \beta_2 BusinessSize + e$$

Applying a Treatment to Sales Example

- You can force the treatment t_0

$$T \leftarrow t_0$$
$$Y \leftarrow f_y(T, u_y)$$

- Ask *"what would happen to the outcome Y if I were to set the treatment to t_0?"*
- The intervention is represented by the $do(.)$ operator, e.g., $do(T = t_0)$
- Note that

$$\mathbb{E}[AmountSold|isOnSales = 1] \neq \mathbb{E}[AmountSold|do(IsOnSales = 1)$$

  - The first is what happens to businesses that make a price cut (e.g., a big business)
    - You are "selecting" the subsample of businesses that actually cut prices
  - The second is that happens if you force every business to engage in sales

SCIENCE ACADEMY

Individual treatment effect

- One can use the $do(.)$ operator to express the impact of the treatment on the outcome for an individual unit i

$$\tau_i = Y_i|do(T = t_1) - Y_i|do(T = t_0)$$

- The effect τ_i of going from treatment t_0 to t_1 for unit i is the difference in the outcome of that unit under t_1 compared to t_0
  - E.g., $AmountSold_i|do(IsOnSales = 1) - AmountSold_i|do(IsOnSales = 0)$
- You can only observe one term due to the fundamental problem of causal inference
  - You can represent it in theory, but you can't recover it from data

Potential outcomes

- Aka "counterfactual"
- The do operator can represent "the unit i outcome would be Y if the treatment is set to t"

$$Y_{t,i} = Y(t)_i = Y_i | do(T_i = t)$$

- If the treatment is binary, you can use $Y_{0,i}$ (without the treatment) vs $Y_{1,i}$ (with the treatment)
  - One is "factual", i.e., observable
  - The other is "counterfactual"

Consistency of the treatment

- There are no hidden multiple versions of the treatment
  - E.g., a drug coming in multiple dosages, but it's treated as binary
  - E.g., in "help from a financial planner", is "help" a one-time consultation or regular advice?

No interference

- Aka "Stable unit of treatment value (SUTVA)"
- The effect of one unit is not influenced by the treatment of other units (if there are spillovers or network effects)
- E.g., vaccinating one person makes other close people less likely to catch the illness even if they don't get the treatment